Name	Equation (if any)	Description
β	$\frac{P}{D}$	Pitch-to-diameter ratio (dimensionless)
β'	$\tan^{-1}\left(\frac{P}{\pi D \bar{r}'}\right)$	Local blade pitch angle at the representative radial station (rad)
σ_e	Geometry-dependent	Effective propeller blade solidity (dimensionless)
J		Advance ratio based on total flight speed
J_{axial}		Axial advance ratio in Pollet's angle convention
J_{T0}, J_{P0}		Zero-Thrust and Zero-Power Advance Ratios
η_T, η_P		Oblique-flow incidence correction factors for thrust and power
δ		Blade-geometry-dependent incidence correction term
C_{T0}, C_{P0}		Static thrust and shaft-power coefficients at hover
$C_{T,P}^{axial}$		Thrust and shaft-power coefficients for axial airflow
C_T^{cor}, C_P^{cor}		Incidence-corrected thrust and shaft-power coefficients for oblique forward flight
$\bar{r}'$		Nondimensional representative radial blade position (typically 0.75)
α		Rotor-axis angle in Pollet's force-equilibrium convention (rad)
θ_{FP}		Flight-path angle relative to the horizontal (rad)
$\phi = \alpha - \theta_{FP}$		Relative angle between the rotor-axis orientation and flight path
S_{top}, S_{front}		Top and frontal projected areas used to estimate aerodynamic reference area (m^2)
C_l, C_d		Airframe lift/downforce and drag coefficients (dimensionless)

There are many ways that can be done to design a quadcopter's propulsion system by simulation or without physical testing, some of them including using factory datasheets or empirical formulas based on existing tested products or to use theoretical estimation based on physics to help to gain important factors that can determine the drone's flying performance such as thrust, power, endurance, etc. In this study, two methods of determining propulsions system will be used on to predict the performance of flying in hover mode and forward mode. The first of them will be to use the theoretical formulas based on physics:

Theoretical hover

$$n = \frac{\omega}{2\pi} = \frac{RPM}{60}$$

Below is the theoretical formula for gaining thrust in Newton and power in Watt:

$$T = C_{T0}\rho n^2 D^4$$

$$P = C_{T0}\rho n^3 D^5$$

If we wanna find RPM needed to hover, we need to first find $T_{hover} = \frac{W}{N_{rotor}}$ which is just the weight of the drone divided by 4:

$$RPM_{hover} = 60\sqrt{\frac{T_{hover}}{(C_{T0}\rho D^4)}}$$

Based on this paper [1], [2] by Pollet and Nam, this formula is empirical for Aeronaut CE 9x5 propeller:

$$C_{T0} = 0.02791 + 0.11867\beta + 0.27334\beta^2 - 0.28852\beta^3$$

$$C_{P0} = 0.01813 - 0.06218\beta + 0.35712\beta^2 - 0.23774\beta^3$$

When we have RPM_{hover} and C_{T0}, C_{P0} its basically done.

Theoretical forward

For theoretical forward, it is a bit complicated if we wanna find the required thrust to fly forward, we start with fbd:

FBD

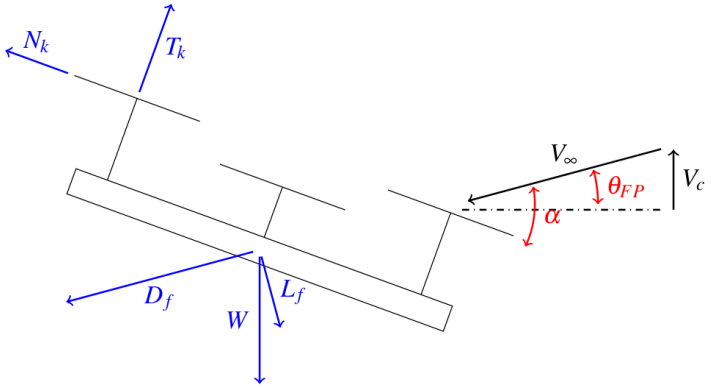
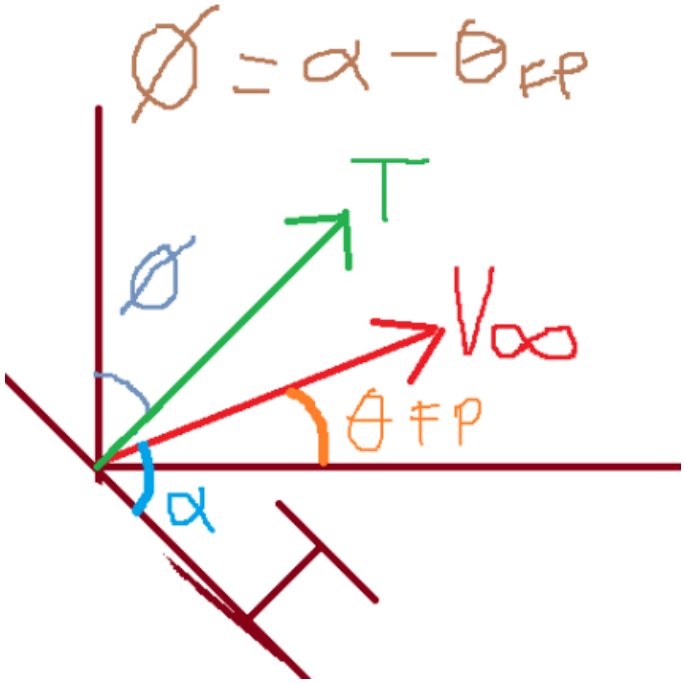


Figure 1 – Quadcopter in Climbing Forward Flight. Side view.



After making the FBD, we use equilibrium eqn. to get these:

$$D_f = \frac{1}{2} C_d \rho_{air} V_\infty^2 S_{ref}$$

$$L_f = \frac{1}{2} C_l \rho_{air} V_\infty^2 S_{ref}$$

$$S_{ref} = S_{top} \sin\alpha + S_{front} \cos\alpha$$

The angle formula

this is from dividing vertical and horizontal equilibrium equations

$$\tan(\alpha - \theta_{FP}) = \frac{D_f \cos\theta_{FP} - L_f \sin\theta_{FP}}{W + D_f \sin\theta_{FP} + L_f \cos\theta_{FP}}$$

$$\tan(\alpha - \theta_{FP}) = \frac{\left(\frac{1}{2} C_d \rho_{air} V_\infty^2 S_{ref}\right) \cos\theta_{FP} - \left(\frac{1}{2} C_l \rho_{air} V_\infty^2 S_{ref}\right) \sin\theta_{FP}}{W + \left(\frac{1}{2} C_d \rho_{air} V_\infty^2 S_{ref}\right) \sin\theta_{FP} + \left(\frac{1}{2} C_l \rho_{air} V_\infty^2 S_{ref}\right) \cos\theta_{FP}}$$

$$\tan(\alpha - \theta_{FP})$$

$$= \frac{\left(\frac{1}{2} C_d \rho_{air} V_\infty^2 \left(S_{top} \sin\alpha + S_{front} \cos\alpha\right)\right) \cos\theta_{FP} - \left(\frac{1}{2} C_l \rho_{air} V_\infty^2 \left(S_{top} \sin\alpha + S_{front} \cos\alpha\right)\right) \sin\theta_{FP}}{W + \left(\frac{1}{2} C_d \rho_{air} V_\infty^2 \left(S_{top} \sin\alpha + S_{front} \cos\alpha\right)\right) \sin\theta_{FP} + \left(\frac{1}{2} C_l \rho_{air} V_\infty^2 \left(S_{top} \sin\alpha + S_{front} \cos\alpha\right)\right) \cos\theta_{FP}}$$

$$\theta_{FP} = 0$$

$$\tan(\alpha) = \frac{\left(\frac{1}{2} C_d \rho_{air} V_\infty^2 \left(S_{top} \sin\alpha + S_{front} \cos\alpha\right)\right)}{W + \left(\frac{1}{2} C_l \rho_{air} V_\infty^2 \left(S_{top} \sin\alpha + S_{front} \cos\alpha\right)\right)}$$

Find alpha that satisfies this for varieties of V_∞, S_{front}, and S_{top}

$$\tan(\alpha) - \frac{\left(\frac{1}{2} C_d \rho_{air} V_\infty^2 \left(S_{top} \sin\alpha + S_{front} \cos\alpha\right)\right)}{W + \left(\frac{1}{2} C_l \rho_{air} V_\infty^2 \left(S_{top} \sin\alpha + S_{front} \cos\alpha\right)\right)} = 0$$

The thrust formula

$$N_{rotors} T = \sqrt{\left(W + D_f \sin\theta_{FP} + L_f \cos\theta_{FP}\right)^2 + \left(D_f \cos\theta_{FP} - L_f \sin\theta_{FP}\right)^2}$$

$$N_{rotors} T = \sqrt{\left(W + \left(\frac{1}{2} C_d \rho_{air} V_\infty^2 S_{ref}\right) \sin\theta_{FP} + \left(\frac{1}{2} C_l \rho_{air} V_\infty^2 S_{ref}\right) \cos\theta_{FP}\right)^2 + \left(\left(\frac{1}{2} C_d \rho_{air} V_\infty^2 S_{ref}\right) \cos\theta_{FP} - \left(\frac{1}{2} C_l \rho_{air} V_\infty^2 S_{ref}\right) \sin\theta_{FP}\right)^2}$$

$$N_{rotors} T = \sqrt{\left(W + \left(\frac{1}{2} C_d \rho_{air} V_\infty^2 \left(S_{top} \sin\alpha + S_{front} \cos\alpha\right)\right) \sin\theta_{FP} + \left(\frac{1}{2} C_l \rho_{air} V_\infty^2 \left(S_{top} \sin\alpha + S_{front} \cos\alpha\right)\right) \cos\theta_{FP}\right)^2 + \left(\frac{1}{2} C_d \rho_{air} V_\infty^2 \left(S_{top} \sin\alpha + S_{front} \cos\alpha\right) \cos\theta_{FP} - \left(\frac{1}{2} C_l \rho_{air} V_\infty^2 \left(S_{top} \sin\alpha + S_{front} \cos\alpha\right)\right) \sin\theta_{FP}\right)^2}$$

$$\theta_{FP} = 0$$

$$T_{forward} = \frac{1}{N_{rotors}} \sqrt{\left(W + \left(\frac{1}{2} C_l \rho_{air} V_\infty^2 \left(S_{top} \sin\alpha + S_{front} \cos\alpha\right)\right)\right)^2 + \left(\frac{1}{2} C_d \rho_{air} V_\infty^2 \left(S_{top} \sin\alpha + S_{front} \cos\alpha\right)\right)^2}$$

Now we can get the RPM forward

$$RPM_{forward} = 60 \sqrt{\frac{T_{forward}}{(C_{T0} \rho D^4)}}$$

So $n(RPM_{forward}) \rightarrow RPM_{forward}(T_{forward}) \rightarrow T_{forward}(\alpha) \rightarrow \alpha(V_\infty,\ C_d,\ C_l,\ S_{top}, S_{front})$

To get thrust and power:

$$T = C_T^{corrected} \rho n^2 D^4$$

$$P = C_P^{corrected} \rho n^3 D^5$$

We then wanna find this:

$$C_{T,P}^{corrected}(\alpha, J) = \eta_{T,P}(\alpha) C_{T,P}(0, J_{axial})$$

Where $\eta_{T,P}(\alpha)$ is incidence correction:

$$\eta_{T,P}(\alpha) = 1 + \frac{\left(\frac{J\cos\alpha}{\pi \bar{r}'^2}\right)^2}{2\left(1 - \frac{J\sin\alpha}{J_{0T,P}^{axial}}\right)}\delta(\alpha)$$

$$\delta(\alpha) = \frac{3}{2}\cos\beta'\left[1 + \left(\frac{\sigma_e}{\tan\beta'}\right)\left(1 + \sqrt{1 + \frac{2\tan\beta'}{\sigma_e}}\right)(1 - \cos\alpha)\right]$$

This is not needed generally but I thought it was part of the formula do I did it anyway

To find $C_{T,P}^{corrected}(\alpha, J)$, the missing components can then be obtained:

$$J = \frac{V_\infty}{nD}$$

$$J_{axial} = J\sin\alpha$$

$$C_T^{axial} = 0.02791 - 0.06543J_{axial} + 0.11867\beta + 0.27334\beta^2 - 0.28852\beta^3 + 0.02104J_{axial}^3 - 0.23504J_{axial}^2 + 0.18677\beta J_{axial}^2$$

$$C_P^{axial} = 0.01813 - 0.06218\beta + 0.00343J_{axial} + 0.35712\beta^2 - 0.23774\beta^3 + 0.07549\beta J_{axial} - 0.1235J_{axial}^2$$

Find $J_{0,T}^{axial}$ and $J_{0,P}^{axial}$:

$$C_T^{axial} = 0.02791 - 0.06543J_{axial} + 0.11867\beta + 0.27334\beta^2 - 0.28852\beta^3 + 0.02104J_{axial}^3 - 0.23504J_{axial}^2 + 0.18677\beta J_{axial}^2 = 0$$

$$0.02104J_{axial}^3 - (0.23504 + 0.18677\beta)J_{axial}^2 - 0.06543J_{axial} + (0.02791 + 0.11867\beta + 0.27334\beta^2 - 0.28852\beta^3) = 0$$

$$C_P^{axial} = 0.01813 - 0.06218\beta + 0.00343J_{axial} + 0.35712\beta^2 - 0.23774\beta^3 + 0.07549\beta J_{axial} - 0.1235J_{axial}^2$$

$$-0.1235J_{axial}^2 + (0.07549\beta + 0.00343)J_{axial} + (0.01813 - 0.06218\beta + 0.35712\beta^2 - 0.23774\beta^3) = 0$$

And then find the minimum real positive root.

Another thing to mention is to use repetition to get the RPM for flying forward correct:

$RPM_1 = 60\sqrt{\frac{T_{forward}}{(C_{T0}\rho D^4)}}$	$C_{T1,P1}^{corrected}(\alpha, J)$

$RPM_2 = 60\sqrt{\frac{T_{forward}}{\left(C_{T1}^{corrected}\rho D^4\right)}}$	$C_{T2,P2}^{corrected}(\alpha, J)$
$RPM_3 = 60\sqrt{\frac{T_{forward}}{\left(C_{T2}^{corrected}\rho D^4\right)}}$	$C_{T3,P3}^{corrected}(\alpha, J)$
…	…

Datasheet method

The same as theoretical but C_T, C_P are obtained from the propeller's datasheet.

Model	Method used to find matched RPM
Theoretical hover	Directly
Theoretical forward	Fixed-point iteration
APC hover	fzero()
APC forward	fzero()

Credit:

F. Pollet, S. Delbecq, M. Budinger, and J.-M. Moschetta, "Design optimization of multirotor drones in forward flight," Sep. 2021.

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S. Delbecq, M. Budinger, A. Ochotorena, A. Reysset, and F. Defay, "Efficient sizing and optimization of multirotor drones based on scaling laws and similarity models," *Aerospace Science and Technology*, vol. 102, p. 105873, Jul. 2020, doi: 10.1016/j.ast.2020.105873.

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